

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 944+620

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

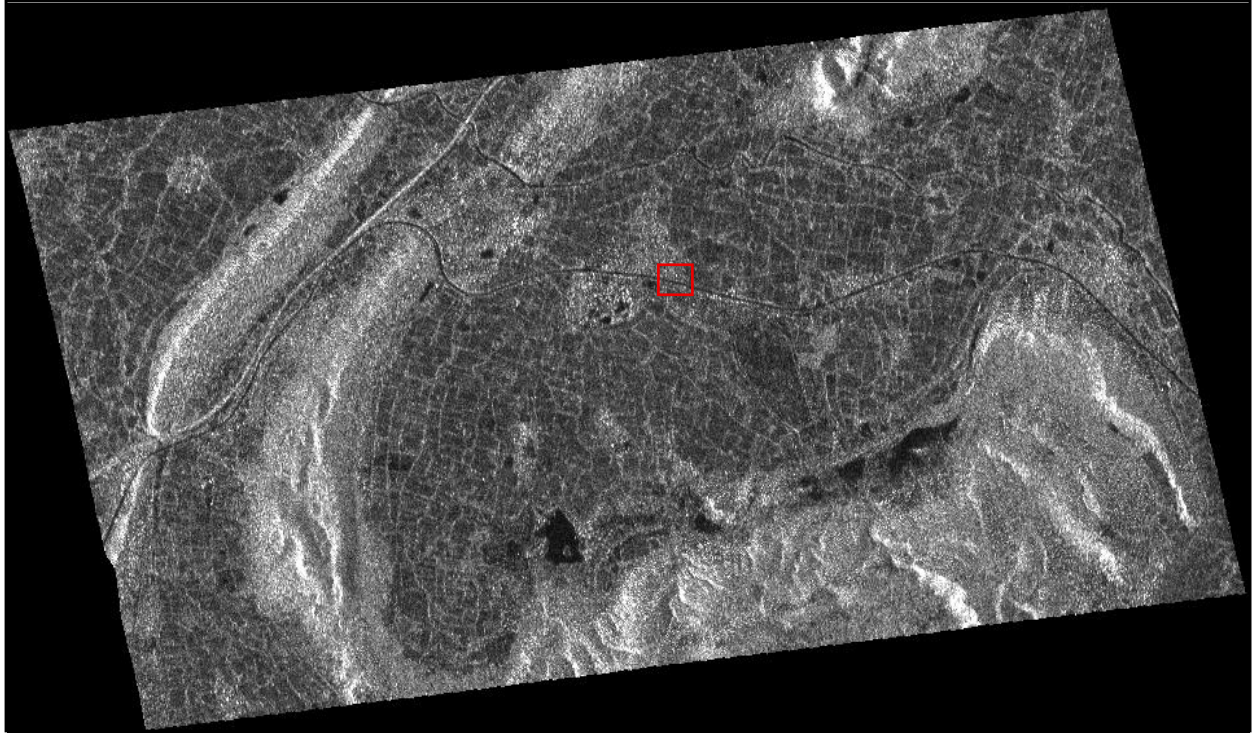
Flight Direction - DESCENDING

Polarization – VV+VH

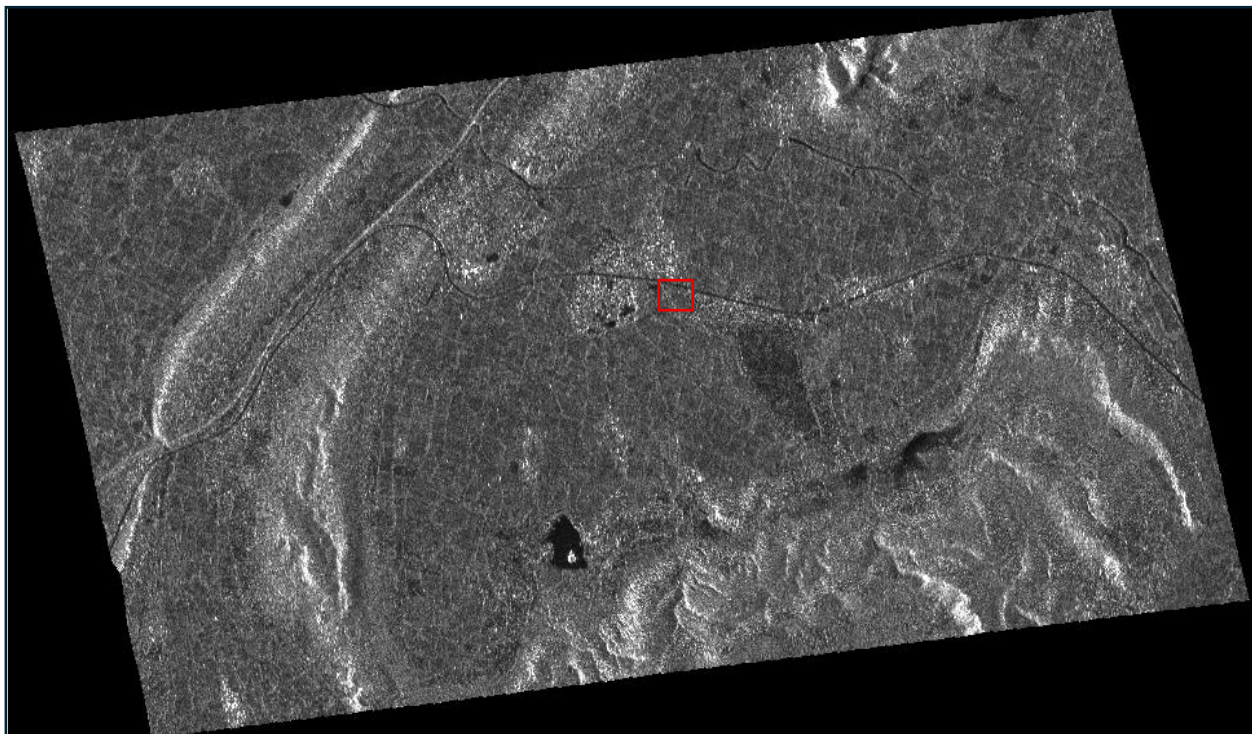
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

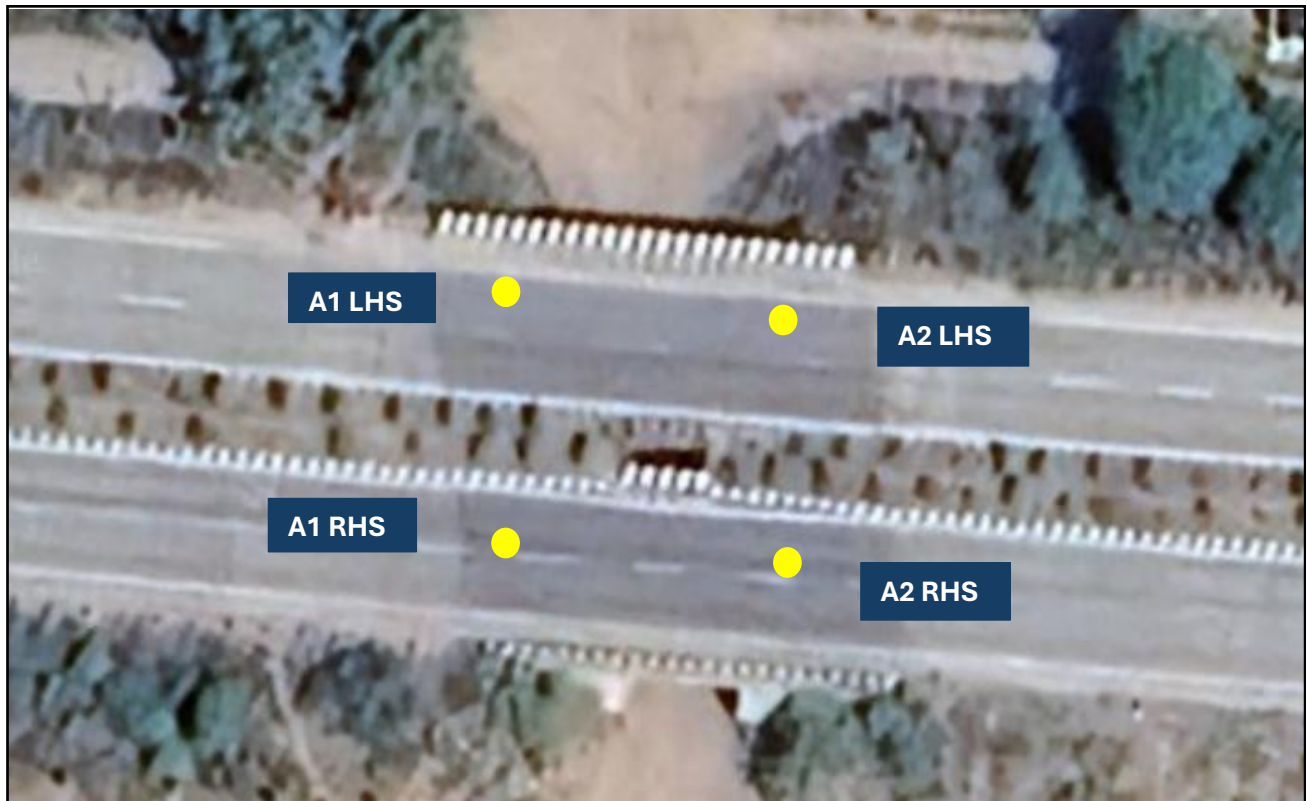


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

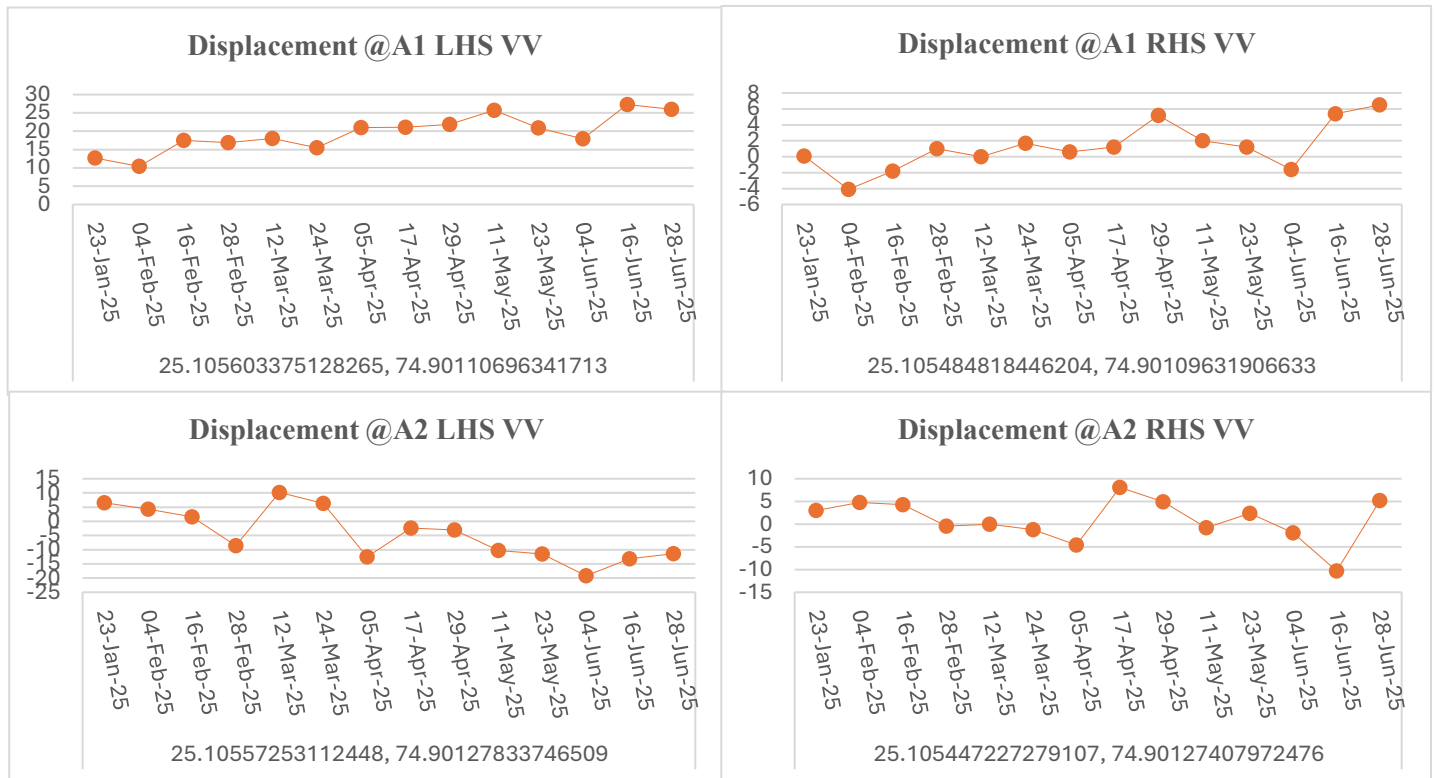
VUP- CH-944+620

Length – 23.03M

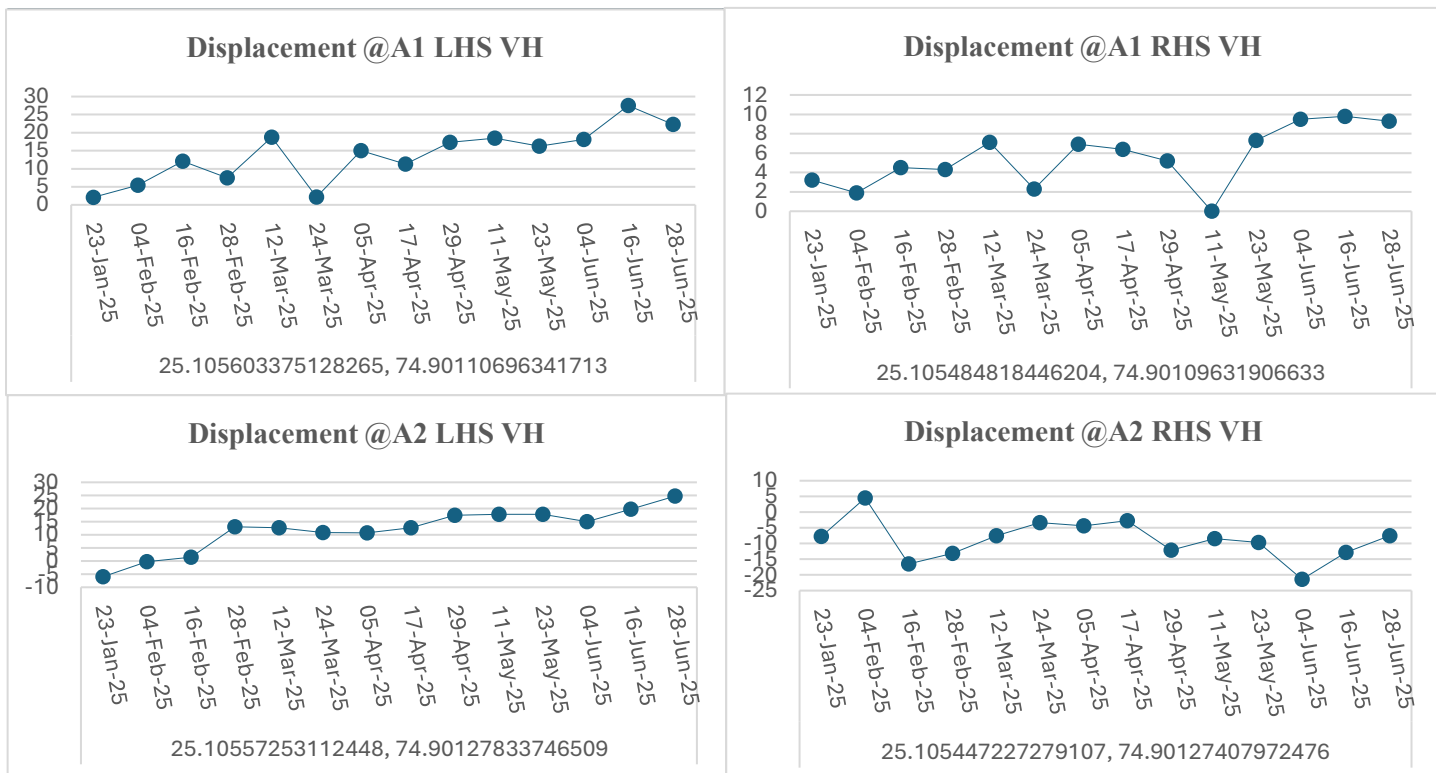


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



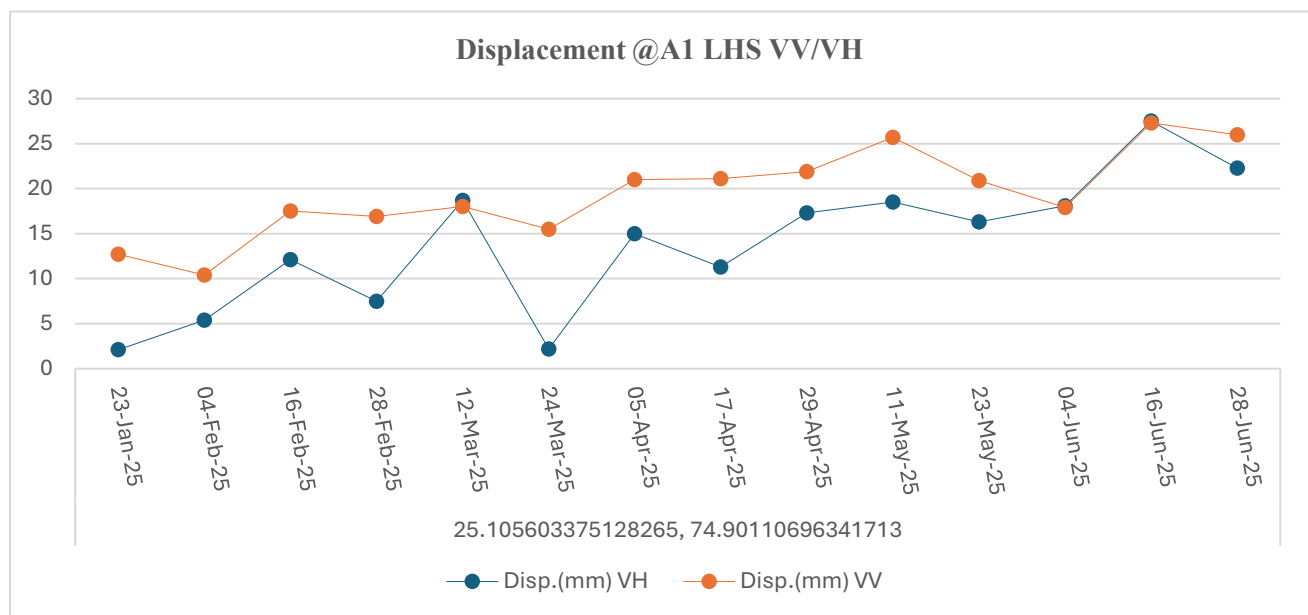
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.105603375128265, 74.90110696341713)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement at A1 LHS exhibits a consistently positive, uplift-dominated behaviour across the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate variability but an overall increasing trend, ranging roughly from +2 mm to +27 mm.

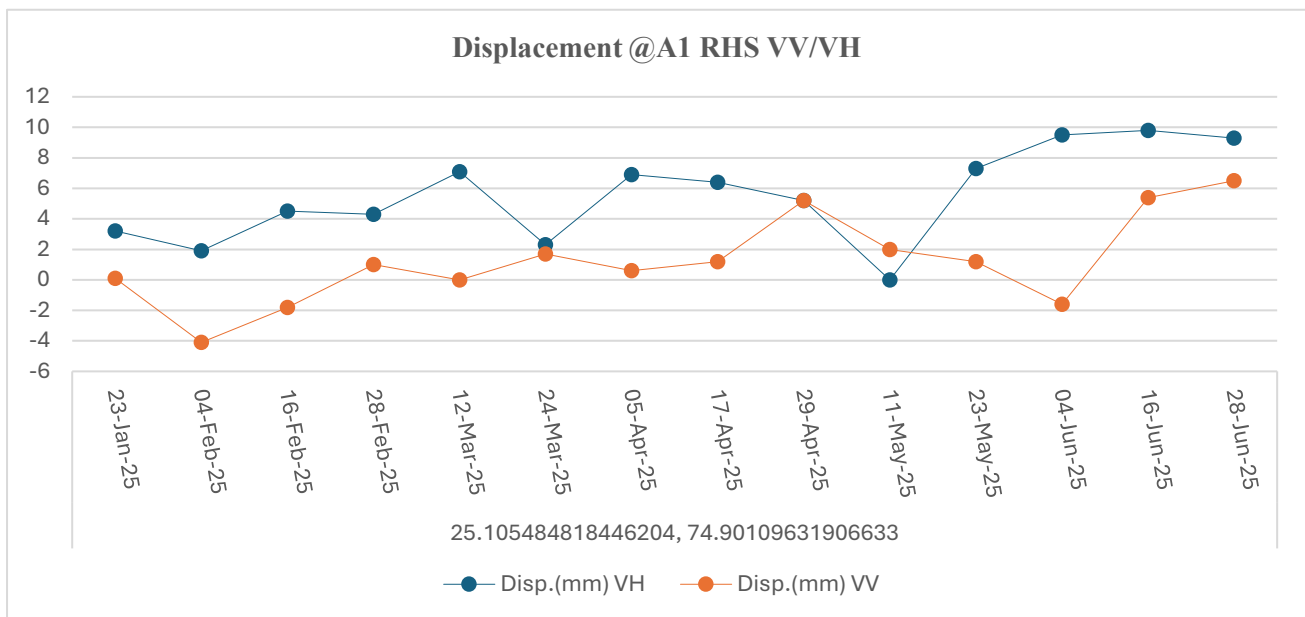
Critical Events & Anomalies:

- **January 2025:** VH begins at +2 mm, while VV is relatively high (+13 mm), indicating initial vertical uplift dominance with limited lateral response.
- **June 2025:** VH reduces slightly to +22 mm, while VV remains high (+26 mm), confirming no structural degradation toward the end of monitoring.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.105484818446204, 74.90109631906633)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows a clear, progressive increasing trend, making VH the dominant deformation component at A1 RHS.
- **Vertical displacement (VV)** Vertical displacement at A1 RHS remains low in magnitude and close to the zero line throughout the monitoring period, indicating minimal vertical deformation.

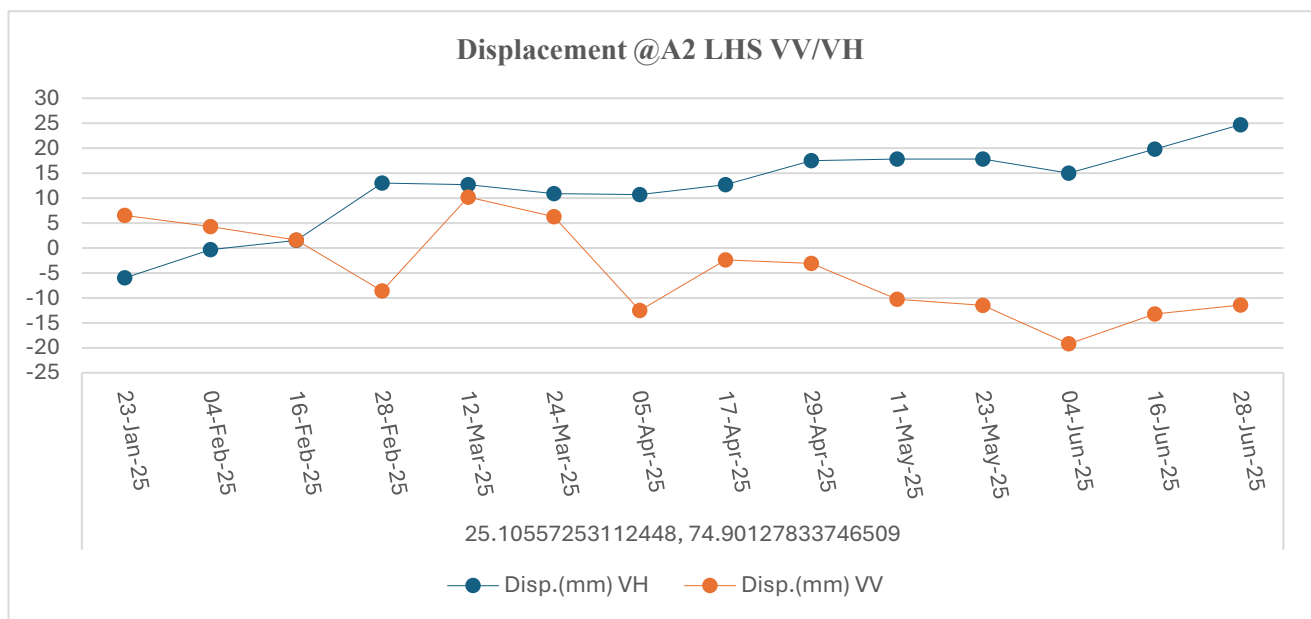
Critical Events & Anomalies:

- **January-February 2025:** VH starts low (+2–3 mm) while VV fluctuates slightly negative (–1 to –4 mm), indicating initial minor vertical settlement with negligible lateral movement.
- **April 2025:** VH dips close to 0 mm, while VV peaks at +5 mm, indicating a short-lived redistribution from lateral to vertical response.
- **June 2025:** VH stabilises near its maximum (+9–10 mm), and VV increases to +6 mm, marking a mild, controlled uplift at the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.10557253112448, 74.90127833746509)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A2 RHS shows moderate to high variability with a clear overall increasing trend.
- **Vertical Displacement (VV)** exhibits predominantly settlement behaviour, with values largely remaining negative throughout the monitoring period.

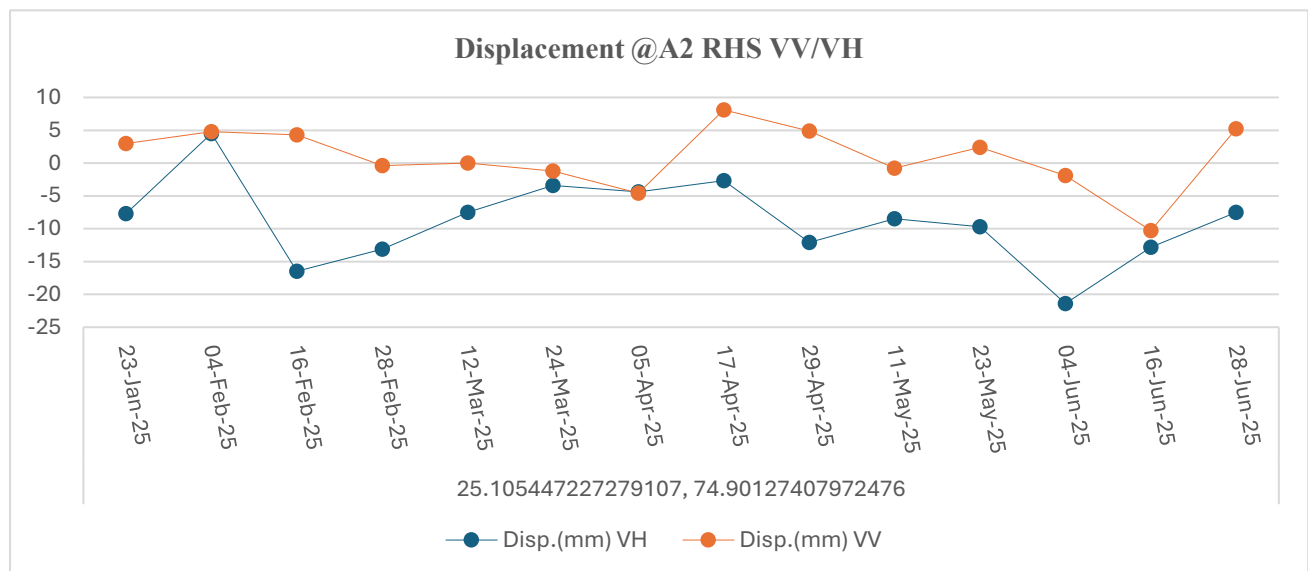
Critical Events & Anomalies:

- **January 2025:** VH initiates near +1 to +3 mm, while VV is already negative (around -5 mm), indicating early vertical settlement preceding significant lateral displacement.
- **June 2025:** VH reaches its maximum value of approximately +27 mm, while VV drops to its lowest magnitude near -25 mm.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.105447227279107, 74.90127407972476)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement shows moderate variability with alternating uplift and settlement, remaining comparatively less extreme than VH.
- **Horizontal (VH) displacement** Horizontal displacement at A2 RHS exhibits pronounced oscillatory behaviour with noticeable fluctuations throughout the monitoring period.

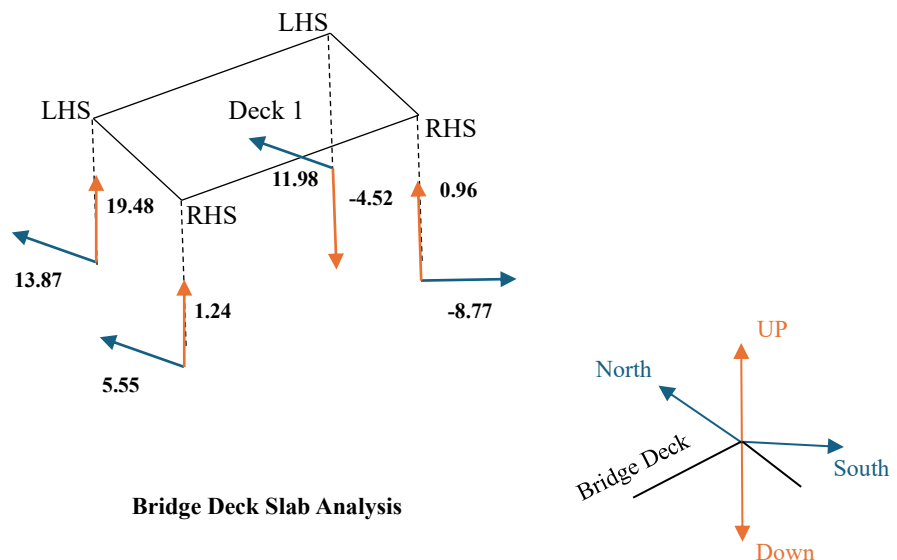
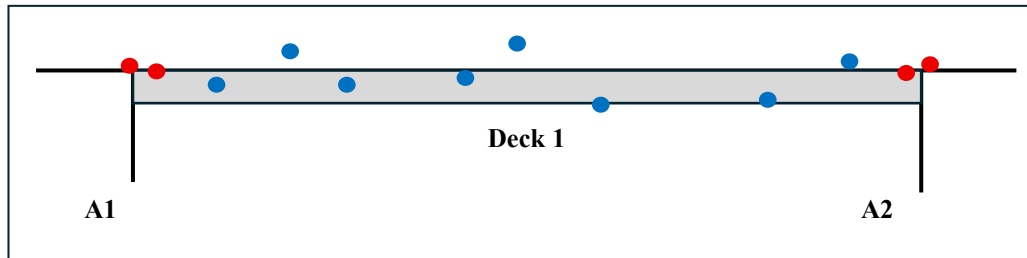
Critical Events & Anomalies:

- **January-February 2025:** VH begins near -7 mm and briefly rises toward +4 mm, while VV remains slightly positive, indicating initial lateral instability with mild vertical uplift.
- **April-May 2025:** Repeated VH troughs between -10 mm and -22 mm coincide with VV oscillating around zero, indicating cyclic lateral deformation possibly influenced by external loading or support condition changes.
- **June 2025:** VH recovers slightly to around -7 mm, while VV rises to +5 mm, suggesting partial structural rebound with stabilized vertical behaviour toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly non-uniform vertical displacement, indicating pronounced differential vertical response between the LHS and RHS regions. The coexistence of large uplift on the LHS and settlement at the deck core confirms strong vertical rotation rather than uniform vertical translation.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates strong directional variability, clearly reflecting torsional deformation of the deck slab. Additional horizontal movement is observed at the LHS lower region (approximately +5.55 mm), reinforcing non-uniform lateral response through the support depth.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by large asymmetric vertical uplift combined with significant lateral displacement. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and support performance.